

NEEL SHAH

Springfield, IL

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Summary

Versatile Data Scientist with 4+ years of experience delivering AI/ML and GenAI solutions across healthcare, agriculture, and technology. Developed LSTM and Transformer models for disease prediction, automated customer support with NLP chatbots, and optimized crop forecasting using deep learning—resulting in a 35% reduction in support time, 25% increase in forecasting accuracy, and 50% boost in operational efficiency. Proficient in Python, SQL, and cloud platforms, with strong expertise in building scalable pipelines and deploying models end-to-end. Created interactive dashboards using Power BI and Tableau to translate insights into strategic decisions. Adept at handling large, complex datasets and integrating structured and unstructured data for actionable outcomes. Recognized for cross-functional collaboration and consistently delivering measurable business value.

Technical Expertise

- **Programming & Development:** Python, SQL, HTML, CSS, JavaScript, Java, PHP, R, VBA, MATLAB, WordPress, Pandas, NumPy, TensorFlow, Keras, PyTorch, Scikit-learn, PySpark, OpenCV, C, C++, Scala
- **Databases & Data Management:** MySQL, MS SQL, MongoDB, Oracle, DB2, SAP, SSRS, SSIS, ETL, Databricks, Apache Hadoop, Apache Spark
- **Data Analysis & Visualization:** Tableau, Power BI, Spotfire, SAS, Qlik, Looker, ggplot2, Matplotlib, Seaborn, Pyplot, Excel, Google Analytics, Google Data Studio
- **Cloud & Big Data:** AWS, Azure, GCP, Docker, Kubernetes, Apache Kafka, Apache Airflow
- **Statistical Analysis & Machine Learning:** Predictive Modeling, Clustering, Decision Trees, Random Forest, XGBoost, A/B Testing, Hypothesis Testing, Time-Series Analysis
- **Methodologies:** CRISP-DM, SEMMA, KDD, Agile, Scrum, Kanban, Lean, Waterfall
- **Business Tools:** Salesforce, Microsoft Office Suite (Excel, Word, PowerPoint, Visio, Access, Project, Teams), ARIS, BPMN, UML
- **DevOps & Reporting Tools:** Jenkins, Prometheus, Grafana, Git, GitHub, GitLab

Professional Experience

Contriva

Feb 2025 - Present

Data Scientist

Plano, Texas

- Engineered AI-driven forecasting models using LSTMs and Transformers to predict global crop demand, improving forecast accuracy by 25%.
- Automated data extraction, transformation, and reporting using GenAI pipelines and Tableau, reducing manual workload by 60%.
- Established ML pipelines using MLflow and Airflow for scalable farm advisory systems, increasing deployment efficiency by 30%.
- Containerized AI applications with Docker and Kubernetes, enhancing cross-environment agricultural analytics and reducing deployment time.
- Analyzed farmer feedback and market sentiment through NLP models using BERT and GPT, enhancing targeted advisories by 30%.
- Designed CNN and U-Net deep learning models to identify high-yield crop variants from imagery, accelerating research timelines by 40%.
- Deployed LLM-powered advisory chatbots for crop rotation and pesticide recommendations, cutting support turnaround time by 35%.
- Linked genomic traits with phenotypes using statistical models and GWAS techniques, supporting development of disease-resistant crops.
- Summarized scientific literature and patents with LLMs for R&D support, saving 15+ hours of manual review weekly.
- Visualized seasonal crop patterns and trends using Power BI and R dashboards, aiding decision-making for farm strategies.
- Collaborated with cross-functional teams to develop GenAI-based yield prediction tools, improving resource allocation by 22%.
- Constructed predictive models in Python and R to assess climate impact on yields, improving planning accuracy by 18%.
- Executed genome sequencing workflows to detect key genetic variants, reducing analysis cycle time by 20%.
- Integrated computer vision with IoT and edge devices for real-time crop monitoring, boosting farm efficiency by 20%.
- Implemented drone-enabled computer vision systems using YOLO, Faster R-CNN, and Mask R-CNN for early disease detection, increasing identification rate by 50%.

Change Healthcare (Optum)

Jan 2024 - Feb 2025

Data Analyst

Nashville, Tennessee

- Collected healthcare data from diverse sources (EMR, claims, clinical notes) and conducted comprehensive data analysis, governance, and visualization using Python and R, improving data accuracy and decision-making processes.
- Utilized ETL framework (Informatica) for seamless transfer and management of patient record data, automating 60% of manual data processes.
- Managed complex healthcare data using Enterprise Data Warehouse (EDW) and data management systems (Teradata, Snowflake), enhancing data accessibility and reliability.
- Demonstrated proficiency in SQL (MySQL) and data analysis tools (Tableau, Power BI) to streamline data collection and visualization, leading to more informed business decisions.

- Applied innovative research methods, including machine learning (ML) algorithms and natural language processing (NLP), to perform predictive analytics and text mining, resulting in improved forecasting and insights.
- Applied medical codes to clinical data, ensuring precision and compliance, achieving a 98% coding accuracy rate.
- Consolidated data from various sources, unveiling hidden patterns crucial for informed decision-making, resulting in a 15% improvement in data-driven insights.
- Demonstrated strong analytical, organizational, and problem-solving skills using SEMMA methodology, contributing to a 10% increase in operational efficiency.
- Created and validated record-keeping processes using in-depth knowledge of healthcare procedures, improving data integrity by 25%.
- Collaborated with insurers, vendors, and healthcare organizations to synthesize crucial data, enabling data-driven decision-making across the healthcare ecosystem, and reducing claims dispute by 15%.

Dynamic Dreamz

Jan 2021 - Jul 2022

Data Analyst

Surat, Gujarat

- Conducted in-depth analysis of user interaction data from web platforms to identify trends and behavioral insights, utilizing Google Analytics and Google Data Studio to enhance decision-making.
- Designed and executed SQL-based data pipelines to extract, transform, and load data from web forms and APIs into structured databases, improving data accessibility by 40%.
- Leveraged Excel, VBA, and Access to automate reporting processes for website usage metrics, reducing manual effort by 50%.
- Built dynamic dashboards using Qlik and Looker, enabling stakeholders to visualize key performance indicators (KPIs) related to user engagement and site performance.
- Applied A/B Testing to assess feature enhancements on user engagement across multiple websites, leading to data-driven UX improvements.
- Employed Hypothesis Testing and Time-Series Analysis to detect anomalies in traffic patterns and predict future site visits.
- Participated in Agile ceremonies including sprint planning and retrospectives, translating analytical requirements into actionable data tasks.
- Documented data collection frameworks and procedures using BPMN and UML, ensuring clarity in data workflows across development and business teams.
- Engaged with cross-functional teams to implement feedback loops from data dashboards, resulting in a 25% improvement in feature adoption rates.
- Adhered to CRISP-DM methodology for structuring analytical approaches and delivering insights aligned with business goals.

Certifications

- Google Data Analytics Professional Certificate
- Data Analytics for Beginners by Google
- Infrastructure and Application Modernization with Google Cloud
- ARIS Business Process Analysis Platform
- HTML, CSS, PHP, JavaScript

Education

University of Illinois at Springfield

Aug 2022 - Dec 2023

Master of Science, Management Information Systems | GPA : 3.8/4.0

USA

- **Coursework:** Data Mining, Information Security, Business Analytics, Database Management Systems (DBMS), System Analysis, Networking, IT Governance, Project Management, Management Information Systems (MIS)

Uka Tarsadia University

Aug 2017 - Jun 2021

Bachelor of Technology, Computer Engineering | GPA : 3.0/4.0

India

- **Coursework:** Data Structures and Algorithms (DSA), Artificial Intelligence and Machine Learning (AI/ML), Object Oriented Programming (OOP), Cloud Computing, Internet of Things (IOT), Web and Mobile Application Development, Operating Systems (OS) and Distributed Computing Systems, Computer Organization and Architecture, Cryptography and Network Security, Business Intelligence (BI)